Final Project Study

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```r
library(readxl)  #for reading excel
library(lmtest)  #for the use of DW test, autocorrelation check
library(forecast)  #for linear reg for time series
library(DIMORA)  #for innovation dif models (GBM, Bass Model etc.)
# library(fpp2) #for datasets
library(naniar)
# library(GGally)
library(ggplot2)
library(corrplot)
library(car)
# library(tseries) library(rcompanion)
library(dplyr)
library(tidyr)
# library(splines)
```

1) Data Exploration & Preprocessing

Load and Inspect the Dataset

```r
ott_cinema_monthly <- read_excel("ott_cinema_dataset.xlsx", sheet = "Monthly")
str(ott_cinema_monthly)
```

```
tibble [216 x 9] (S3: tbl_df/tbl/data.frame)
 $ Time                    : chr [1:216] "2006-01" "2006-02" "2006-03" "2006-04" ...
 $ cinema_admissions       : num [1:216] 13700979 8207270 10102967 10872037 9408449 ...
 $ cinema_screenings       : num [1:216] 120005 102136 109564 110636 103219 ...
 $ cinema_audience_spending: num [1:216] 89522221 50597282 61407884 68866213 58029889 ...
 $ cinema_avg_spending_pp  : num [1:216] 6.53 6.16 6.08 6.33 6.17 ...
 $ gooogle_trend_ott       : num [1:216] 0 0 0 0 0 0 0 0 0 0 ...
 $ cpi_all                 : num [1:216] 0.2 0.2 0.2 0.3 0.2 0.2 0.4 0.2 -0.1 -0.2 ...
 $ cpi_culture             : num [1:216] 0.6 -0.2 0.2 -0.2 -0.1 0.1 0.4 0.7 0.2 -0.6 ...
 $ covid_cases             : num [1:216] 0 0 0 0 0 0 0 0 0 0 ...
```

```r
ott_cinema_annual <- read_excel("ott_cinema_dataset.xlsx", sheet = "Annual")
str(ott_cinema_annual)
```

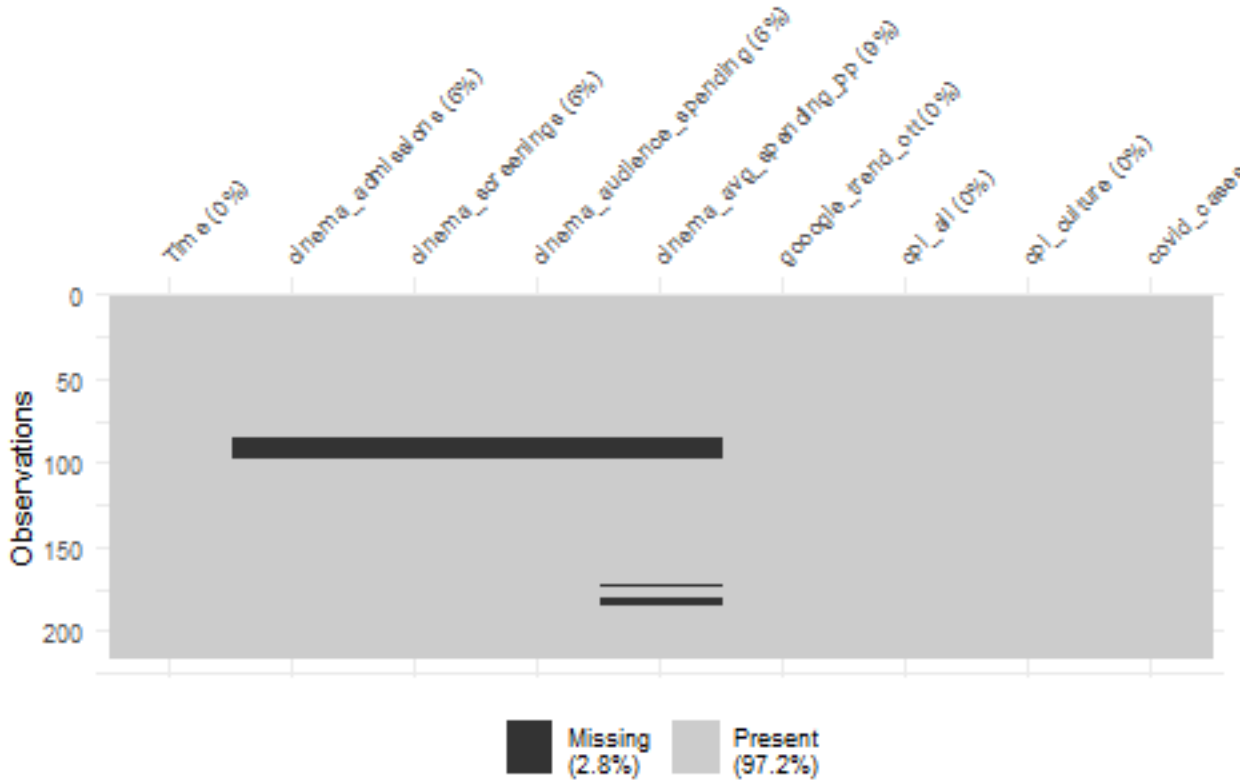
```
tibble [18 x 4] (S3: tbl_df/tbl/data.frame)
 $ Year               : num [1:18] 2006 2007 2008 2009 2010 ...
 $ ott_subscriptions_M: num [1:18] 0 0 0 0 0 0 0 0 0.3 0.8 ...
 $ ott_spending_M_EUR : num [1:18] 0 0 0.6 0.7 1.4 9.5 21.9 21.1 39 64.1 ...
 $ cinema_admissions  : num [1:18] 1.06e+08 1.17e+08 1.12e+08 1.11e+08 1.23e+08 ...
```

```r
ott_cinema_monthly$Time = as.Date(paste0(ott_cinema_monthly$Time, "-01"),
    format = "%Y-%m-%d")
# ott_cinema_annual$Year <- as.Date(paste0(ott_cinema_annual$Year,
# '-01-01'), format='%Y-%m-%d')
```

```r
# Check for missing values
ott_cinema_monthly[apply(is.na(ott_cinema_monthly), 1, any), ]
```

```
# A tibble: 19 x 9
   Time       cinema_admissions cinema_screenings cinema_audience_spending
   <date>                 <dbl>             <dbl>                     <dbl>
 1 2013-01-01                NA                NA                        NA
 2 2013-02-01                NA                NA                        NA
 3 2013-03-01                NA                NA                        NA
 4 2013-04-01                NA                NA                        NA
 5 2013-05-01                NA                NA                        NA
 6 2013-06-01                NA                NA                        NA
 7 2013-07-01                NA                NA                        NA
 8 2013-08-01                NA                NA                        NA
 9 2013-09-01                NA                NA                        NA
10 2013-10-01                NA                NA                        NA
11 2013-11-01                NA                NA                        NA
12 2013-12-01                NA                NA                        NA
13 2020-04-01                 0                 0                       326.
14 2020-05-01                 0                 0                       198
15 2020-11-01                 0                 0                      4824.
16 2020-12-01                 0                 0                     56332
17 2021-01-01                 0                 0                     27483
18 2021-02-01                 0                 0                     14354.
19 2021-03-01                 0                 0                      2165.
# i 5 more variables: cinema_avg_spending_pp <dbl>, gooogle_trend_ott <dbl>,
#   cpi_all <dbl>, cpi_culture <dbl>, covid_cases <dbl>
```

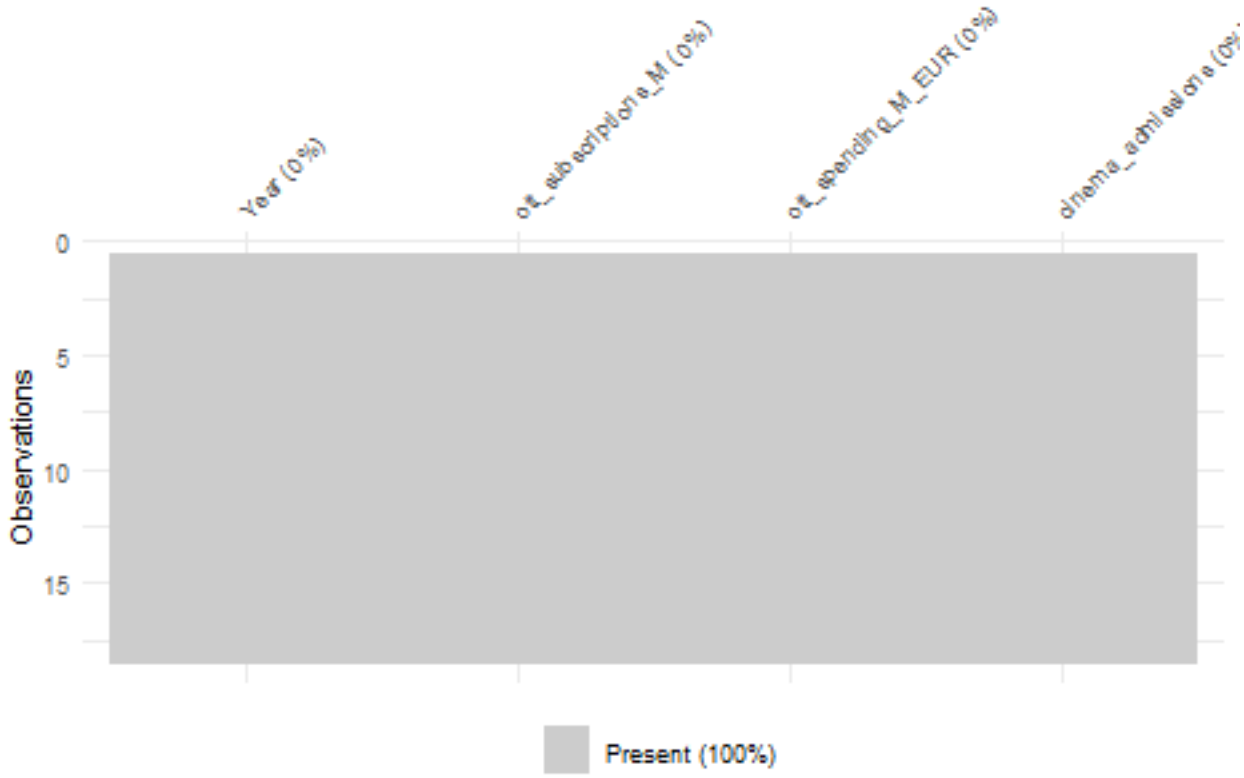
```r
vis_miss(ott_cinema_monthly)
```




```
ott_cinema_annual[apply(is.na(ott_cinema_annual), 1, any), ]
```

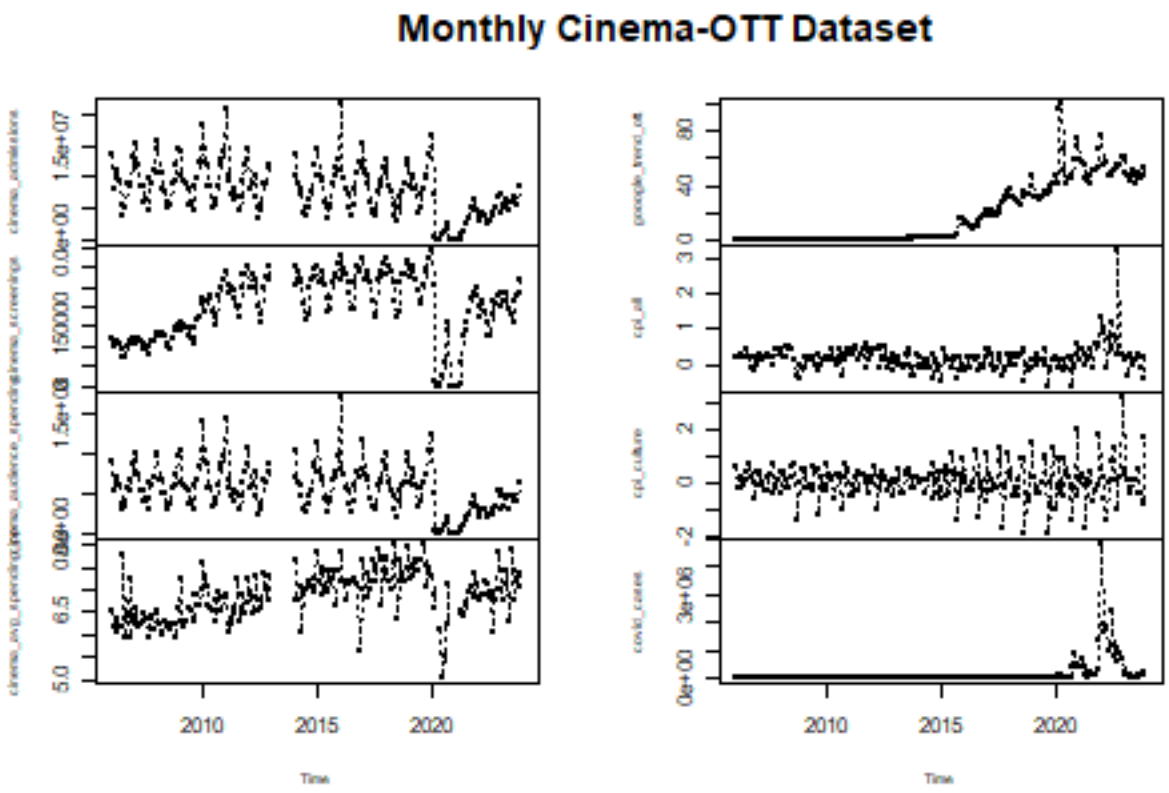
```
# A tibble: 0 x 4
# i 4 variables: Year <dbl>, ott_subscriptions_M <dbl>,
#   ott_spending_M_EUR <dbl>, cinema_admissions <dbl>
```

```
vis_miss(ott_cinema_annual)
```

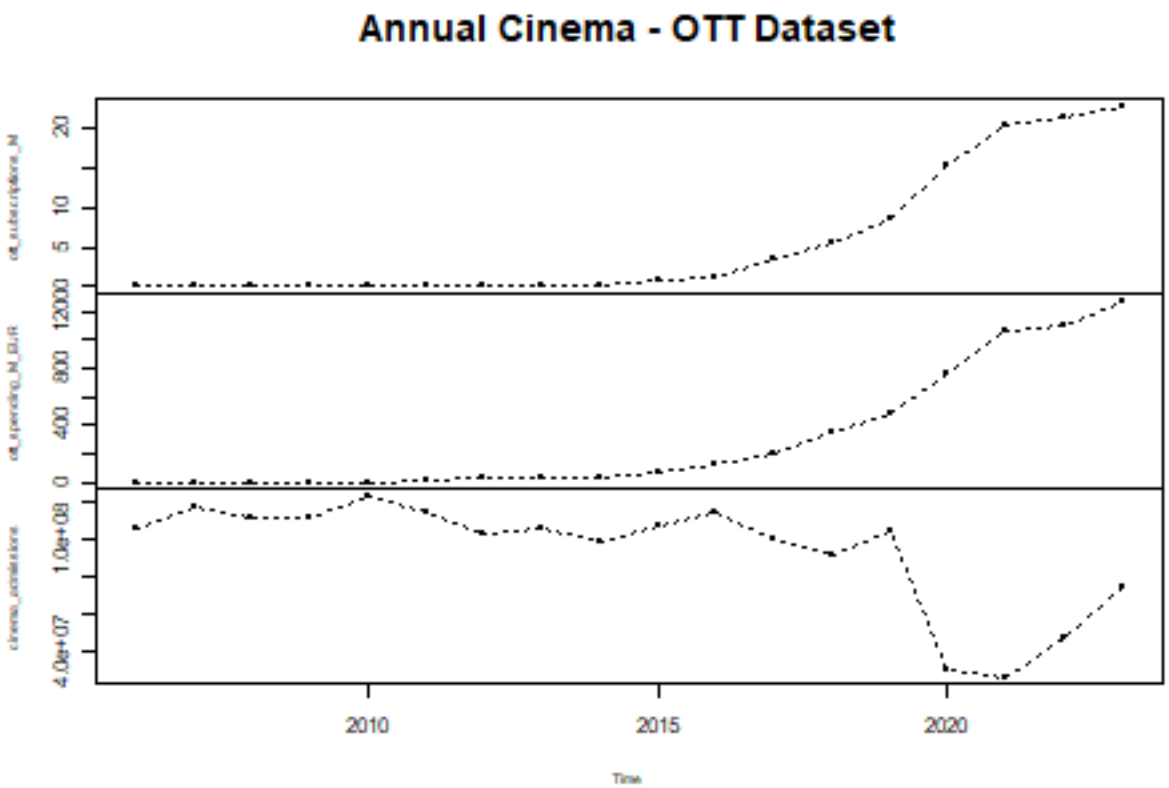



```r
ott_cinema_monthly.ts <- ts(ott_cinema_monthly[, -1], start = c(2006, 1),
    end = c(2023, 12), frequency = 12)
ott_cinema_annual.ts <- ts(ott_cinema_annual[, -1], start = c(2006), end = c(2023),
    frequency = 1)
```

```r
plot(ott_cinema_monthly.ts, main = "Monthly Cinema-OTT Dataset", cex.lab = 0.45,
    type = "o", pch = 16, lty = 3, cex = 0.6)
```




```
plot(ott_cinema_annual.ts, main = "Annual Cinema - OTT Dataset", cex.lab = 0.45,
    type = "o", pch = 16, lty = 3, cex = 0.6)
```




```
# read the data cinema<- read_excel('cinema_monthly.xlsx') str(cinema)

# cinema.ts <- ts(cinema[,-1], start=c(2006,1), end=c(2023,12),
# frequency = 12)
```

Handle Missing Values

Handle Monthly Dataset Missing Values

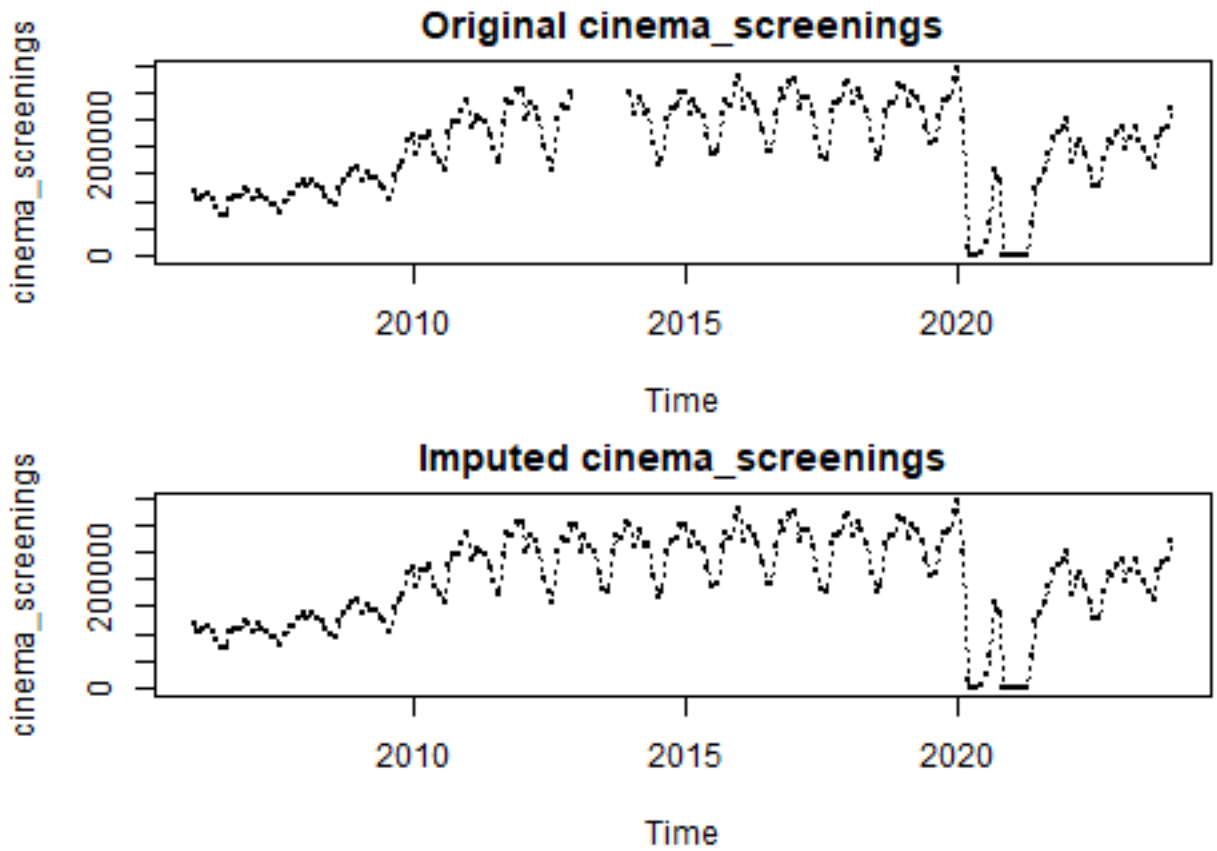
```
# Define the total annual values for 2013 for each column
total_2013 <- list(cinema_screenings = 3014642, cinema_admissions = 106124433,
    cinema_audience_spending = 732299026.82)

# Define the columns to impute
columns_to_impute <- c("cinema_screenings", "cinema_admissions", "cinema_audience_spending",
    "cinema_avg_spending_pp")

# Create a copy of the time series object for imputation
cinema_imputed <- ott_cinema_monthly.ts

# Filter data to include only 2011, 2012, 2014, 2015 for monthly
# averages
selected_years <- (time(ott_cinema_monthly.ts) >= 2011 & time(ott_cinema_monthly.ts) <
```

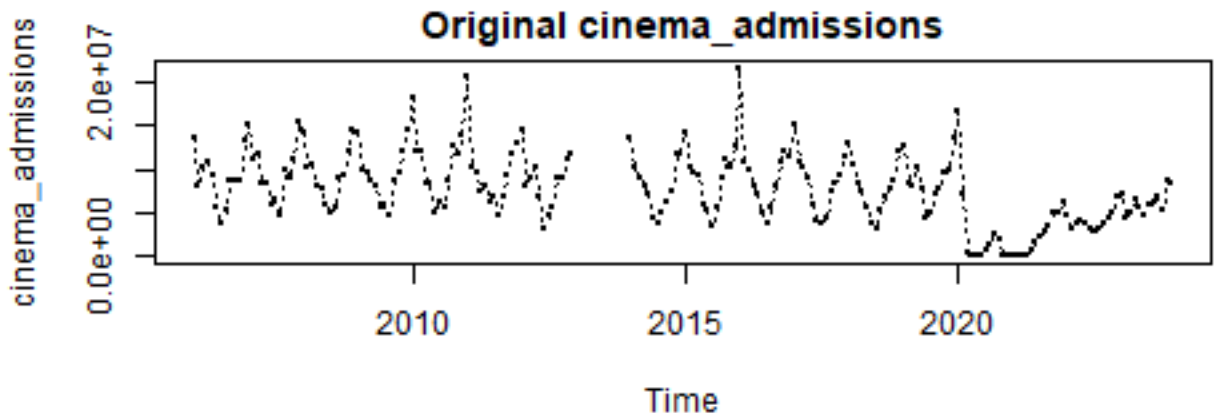
```r
    2013) | (time(ott_cinema_monthly.ts) >= 2014 & time(ott_cinema_monthly.ts) <
    2016)
```

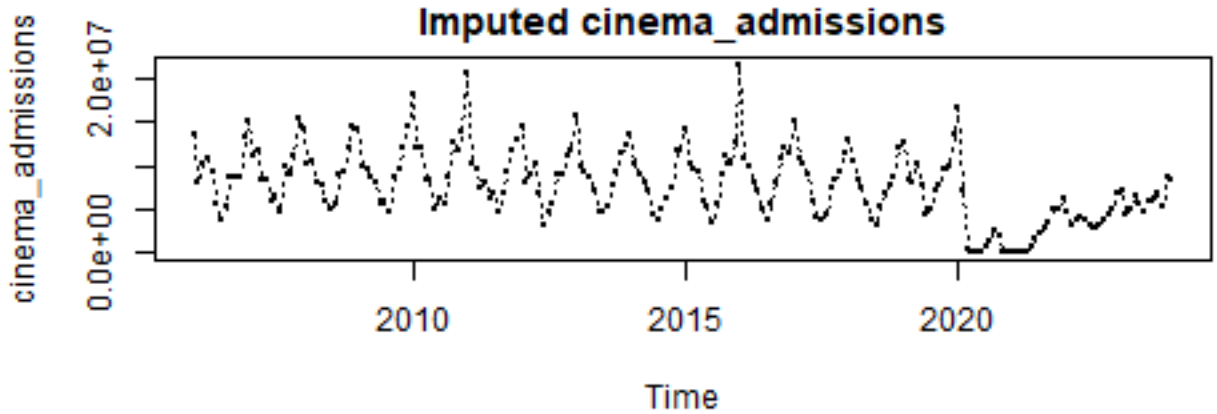
```r
# Loop through each column to impute missing values
for (col in columns_to_impute) {
    # Extract the column data
    column_data <- as.numeric(ott_cinema_monthly.ts[, col])

    if (col == "cinema_avg_spending_pp") {
        # Calculate avg_spending_pp using the imputed
        # audience_spending/audience_admissions
        imputed_2013 <- cinema_imputed[, "cinema_audience_spending"]/cinema_imputed[,
            "cinema_admissions"]

        # Replace the missing 2013 data with the calculated values
        column_data[time(ott_cinema_monthly.ts) >= 2013 & time(ott_cinema_monthly.ts) <
            2014] <- imputed_2013[time(ott_cinema_monthly.ts) >= 2013 & time(ott_cinema_monthly.ts) <
            2014]

        # Handle missing values in 2020 and 2021 (due to COVID-19
        # impact)
        column_data[time(ott_cinema_monthly.ts) >= 2020 & time(ott_cinema_monthly.ts) <
            2021 & is.na(column_data)] <- 6.35  # Average for 2020
        column_data[time(ott_cinema_monthly.ts) >= 2021 & time(ott_cinema_monthly.ts) <
            2022 & is.na(column_data)] <- 6.76  # Average for 2021
    } else {
        # Handle cumulative features using selected years for
        # proportions
        monthly_proportions <- colMeans(matrix(column_data[selected_years],
            ncol = 12, byrow = TRUE), na.rm = TRUE)
        # Normalize to proportions
        monthly_proportions <- monthly_proportions/sum(monthly_proportions)
        imputed_2013 <- monthly_proportions * total_2013[[col]]

        # Replace the missing 2013 data with the imputed values
        column_data[time(ott_cinema_monthly.ts) >= 2013 & time(ott_cinema_monthly.ts) <
            2014] <- imputed_2013
    }

    # Update the imputed data in the time series
    cinema_imputed[, col] <- column_data
}
```

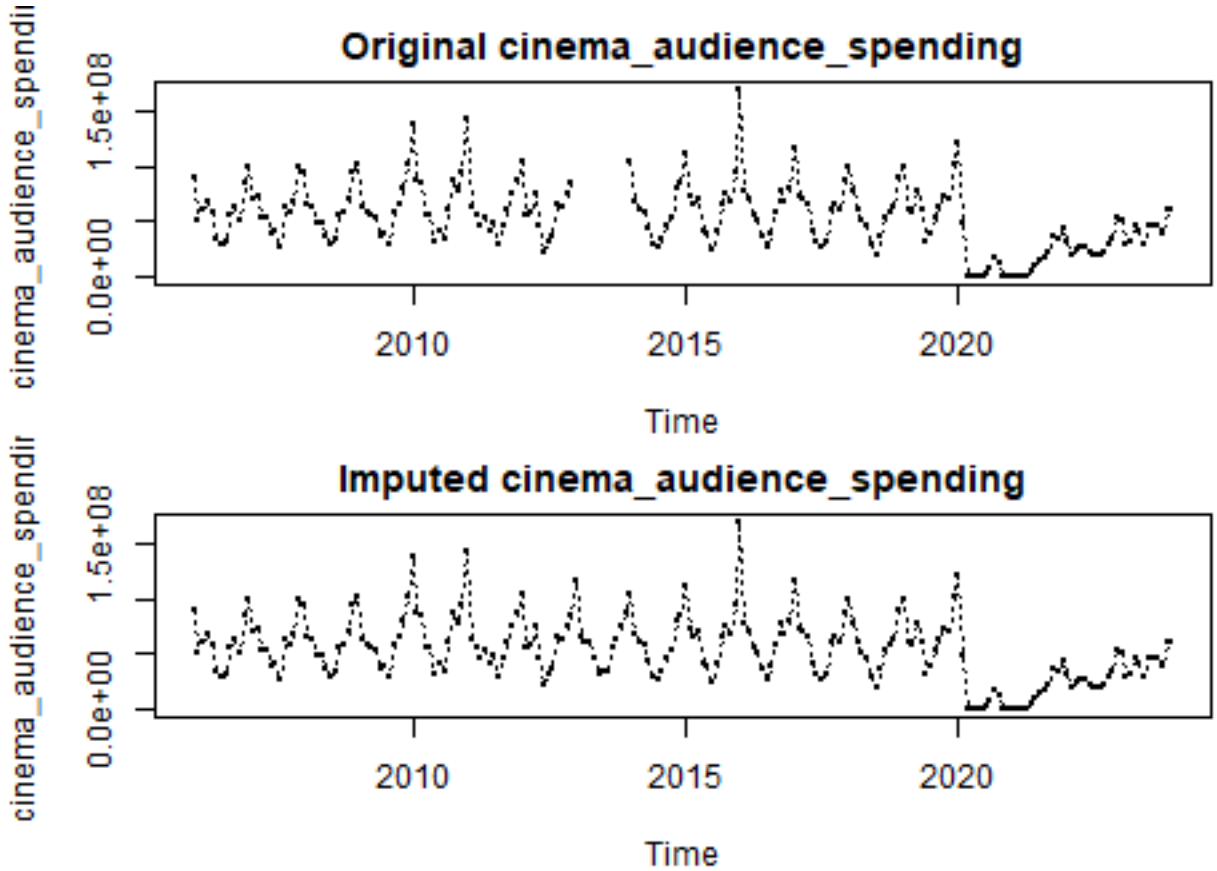
```r
# Plot the original and imputed data for comparison
for (col in columns_to_impute) {
    par(mfrow = c(2, 1), mar = c(4, 4, 2, 1))

    # Plot the original data
    plot(ott_cinema_monthly.ts[, col], type = "o", pch = 16, lty = 3, cex = 0.4,
        main = paste("Original", col), xlab = "Time", ylab = col)

    # Plot the imputed data
    plot(cinema_imputed[, col], type = "o", pch = 16, lty = 3, cex = 0.4,
```

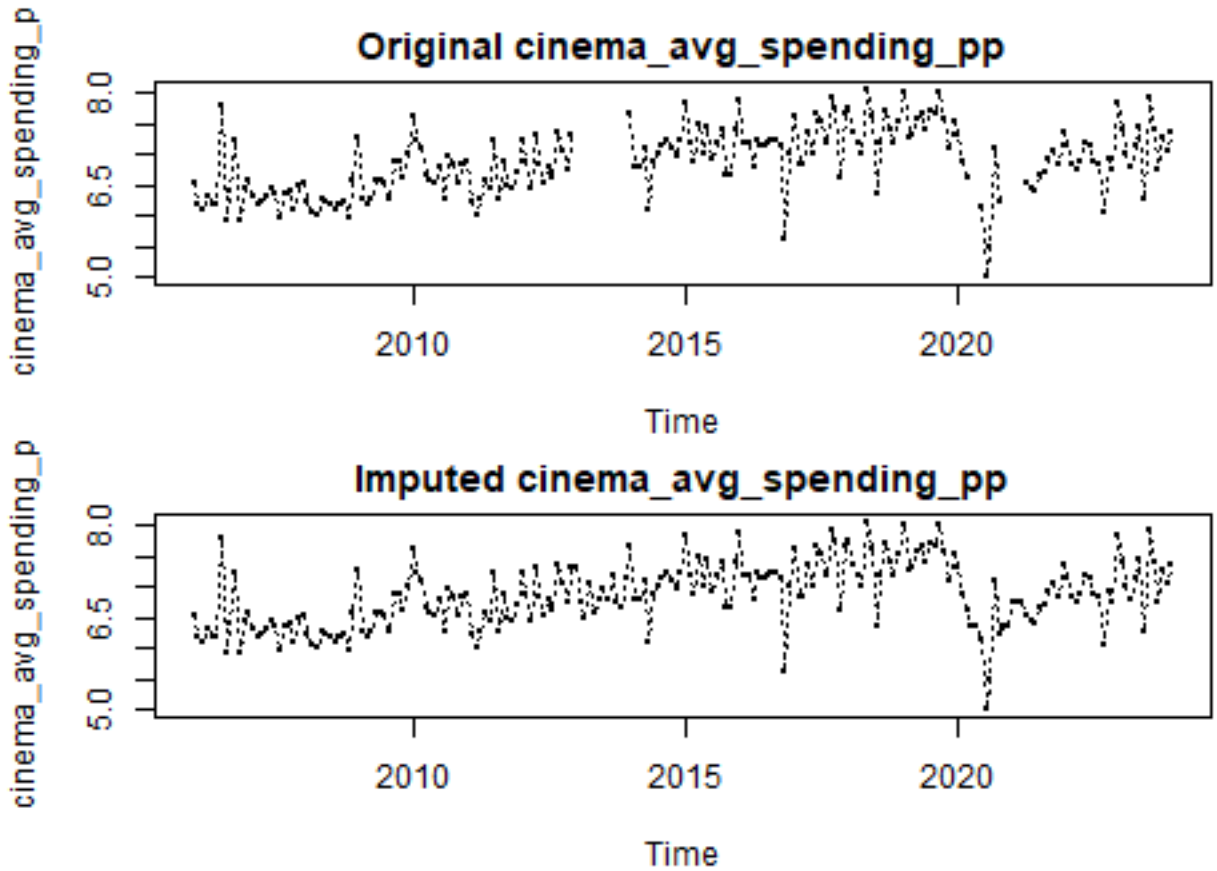
```
        main = paste("Imputed", col), xlab = "Time", ylab = col)
}
```




Original cinema_admissions
cinema_admissions
0.0e+00
2.0e+07
2010
2015
2020
Time

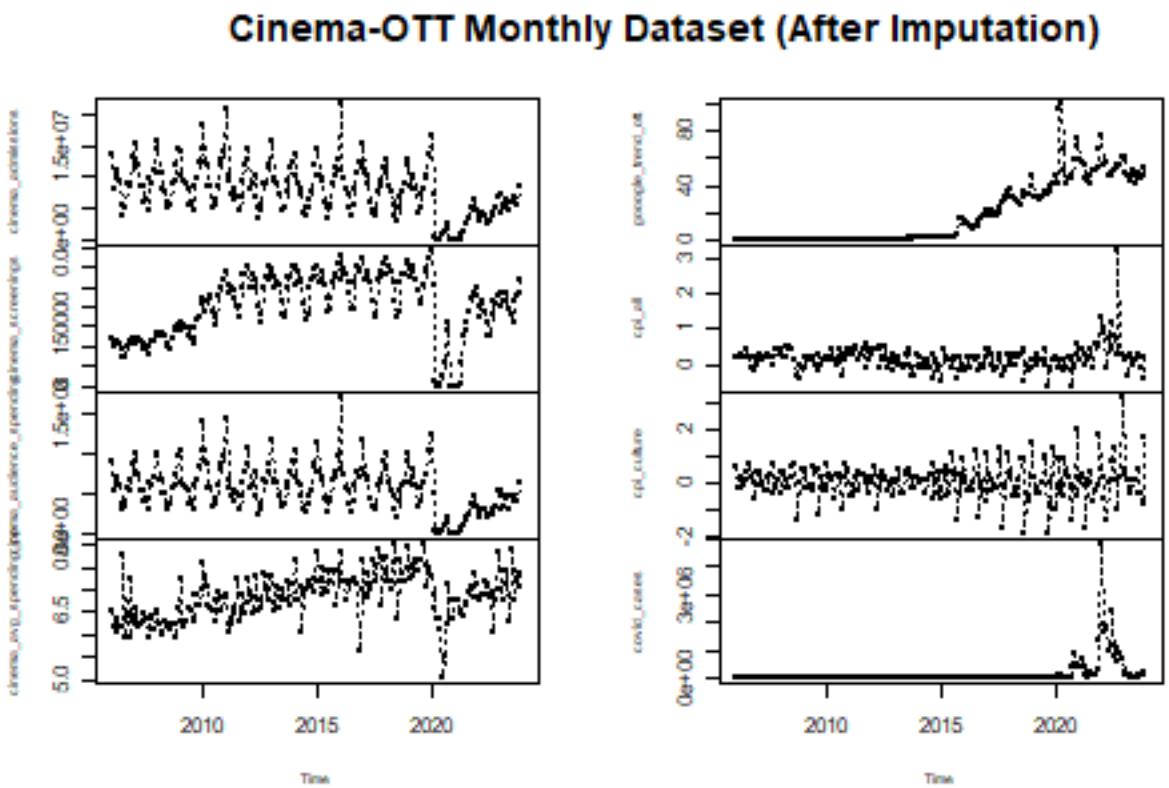

Imputed cinema_admissions
cinema_admissions
0.0e+00
2.0e+07
2010
2015
2020
Time


Original cinema_audience_spending
cinema_audience_spendi
0.0e+00
1.5e+08
2010
2015
2020
Time
Imputed cinema_audience_spending
cinema_audience_spendir
0.0e+00
1.5e+08
2010
2015
2020
Time




```r
par(mfrow = c(1, 1))
```

```r
ott_cinema_monthly.ts <- cinema_imputed
# Final plot of the imputed dataset
plot(ott_cinema_monthly.ts, main = "Cinema-OTT Monthly Dataset (After Imputation)",
    cex.lab = 0.45, type = "o", pch = 16, lty = 3, cex = 0.6)
```



Handle Annual Dataset Missing Values

```
# Extract the households_w_internet column households_ts <-
# ott_cinema_annual.ts[,9]

# Fit a Damped Holt's Exponential Smoothing Model holt_model <-
# holt(households_ts, damped = TRUE, h = 1)

# Forecast the missing value for 2023 forecast_2023 <-
# forecast(holt_model, h = 1)$mean

# Fill the missing value in the dataset
# ott_cinema_annual$households_w_internet[which(is.na(ott_cinema_annual$households_w_internet))]
# <- forecast_2023

# ott_cinema_annual.ts <- ts(ott_cinema_annual[,-1], start=c(2006),
# end=c(2023), frequency = 1)

# The updated dataset tail(ott_cinema_annual[,c(1,10)])
```

```
# Plot the fitted model and forecast autoplot(holt_model) +
# autolayer(fitted(holt_model), series = 'Fitted Values', color =
# 'blue') + labs(title = 'Damped Holt's Exponential Smoothing for
```

```
# Households with Internet', x = 'Year', y = '% Households with
# Internet')
```

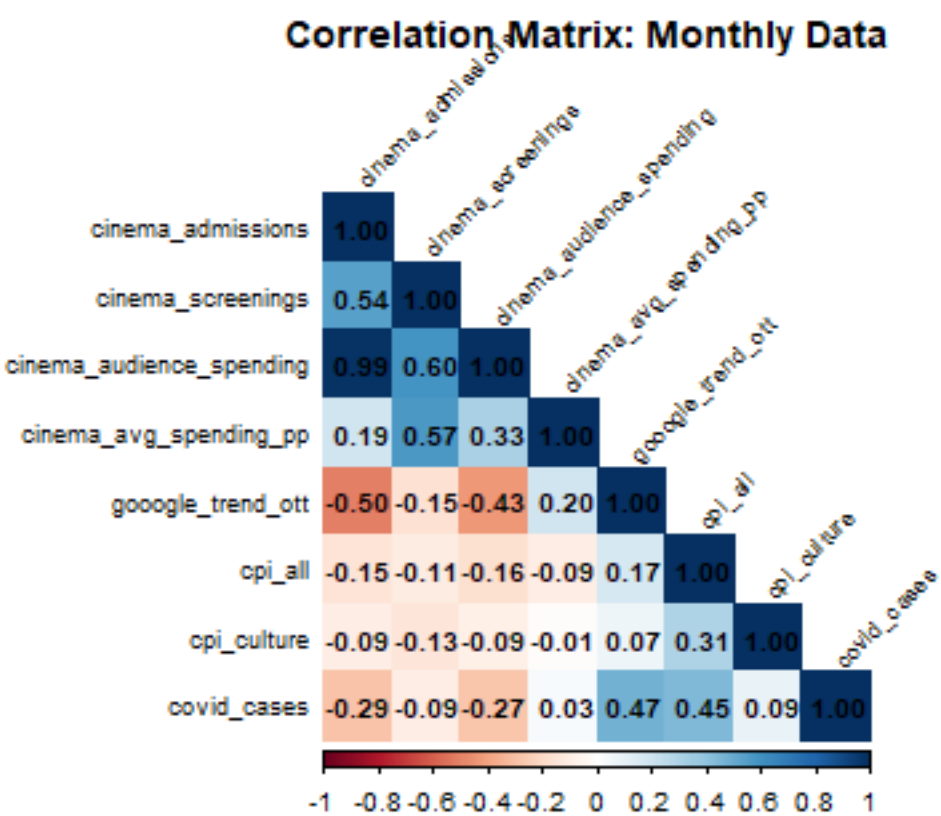
```
# plot(ott_cinema_annual.ts, main='Annual Cinema-OTT Dataset (After
# Imputation)', cex.lab=0.45, type='o', pch = 16, lty = 3, cex=0.6)
```

Collinearity Check

Correlation Matrix for Monthly Data

```
# Correlation Matrix for Monthly Data
ott_cinema_monthly <- as.data.frame(ott_cinema_monthly.ts)

# Compute correlation matrix
cor_matrix_monthly <- cor(ott_cinema_monthly, use = "complete.obs")
```

```
# default_margins <- par('mar') par(mar = c(5, 5, 4, 2) + 0.1)

# Visualize correlation matrix
corrplot(cor_matrix_monthly, method = "color", type = "lower", tl.col = "black",
    tl.srt = 45, addCoef.col = "black", number.cex = 0.8, tl.cex = 0.75)
title(main = "Correlation Matrix: Monthly Data", line = 2, cex.main = 1.2)
```



```r
# par(mar = default_margins)
```

VIF Analysis for Monthly Data

```r
# Run VIF analysis
vif_model.monthly <- lm(cinema_admissions ~ . - cinema_admissions, data = ott_cinema_monthly)  # Regres
vif.monthly <- vif(vif_model.monthly)  # Compute VIF

vif.monthly
```

```
       cinema_screenings cinema_audience_spending   cinema_avg_spending_pp
                2.126995                 1.976895                 1.771816
       gooogle_trend_ott                  cpi_all              cpi_culture
                1.746757                 1.389224                 1.130118
             covid_cases
                1.578432
```

```r
# Compute correlation matrix cor_matrix_annual <-
# cor(ott_cinema_annual[,-1], use = 'complete.obs')

# Visualize correlation matrix corrplot(cor_matrix_annual, method =
# 'color', type = 'lower', tl.col = 'black', tl.srt = 45, addCoef.col =
# 'black', number.cex = 0.65, , tl.cex=0.55) title(main = 'Correlation
# Matrix: Annual Data', line = 2, cex.main = 1.2)
```

Check correlation & trends between OTT Data (Annual) and Google Trends OTT (Monthly)

```r
ott_cinema_monthly$Year <- rep(2006:2023, each = 12)  # Create a Year column
```

```r
# Aggregate Google Trends OTT monthly data to annual (sum)
google_trends_annual <- ott_cinema_monthly %>%
    group_by(Year) %>%
    summarise(google_trend_ott = sum(gooogle_trend_ott, na.rm = TRUE))
```

```r
# Merge with annual dataset
merged_ott_data <- merge(ott_cinema_annual[, c(1:3)], google_trends_annual,
    by.x = "Year", by.y = "Year", all.x = TRUE)
```

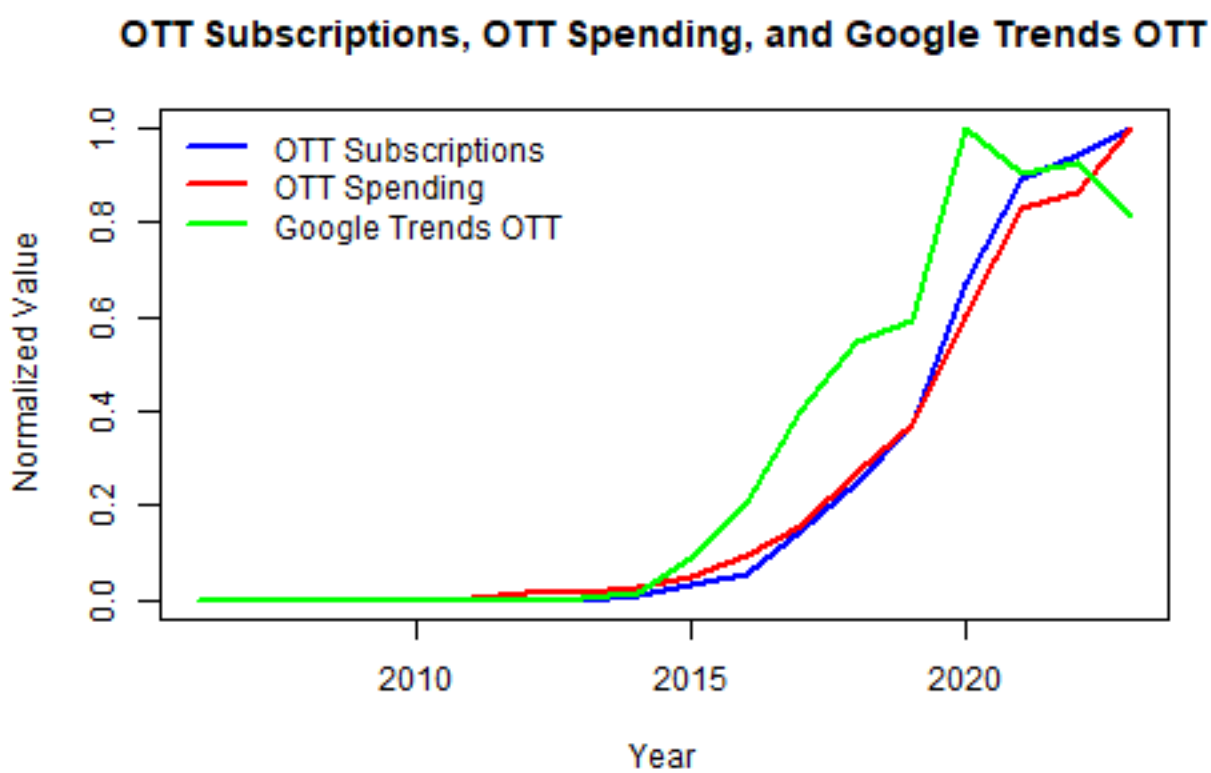
```r
# Compute Correlation Matrix
cor_matrix <- cor(merged_ott_data[, -1], use = "complete.obs")
cor_matrix
```

```
                    ott_subscriptions_M ott_spending_M_EUR google_trend_ott
ott_subscriptions_M           1.0000000          0.9980075        0.9379704
ott_spending_M_EUR            0.9980075          1.0000000        0.9344162
google_trend_ott              0.9379704          0.9344162        1.0000000
```

```r
# Visualize Correlation Matrix corrplot(cor_matrix, method = 'color',
# type = 'lower', tl.col = 'black', tl.srt = 45, addCoef.col = 'black',
# number.cex = 0.8, tl.cex=0.65)
```

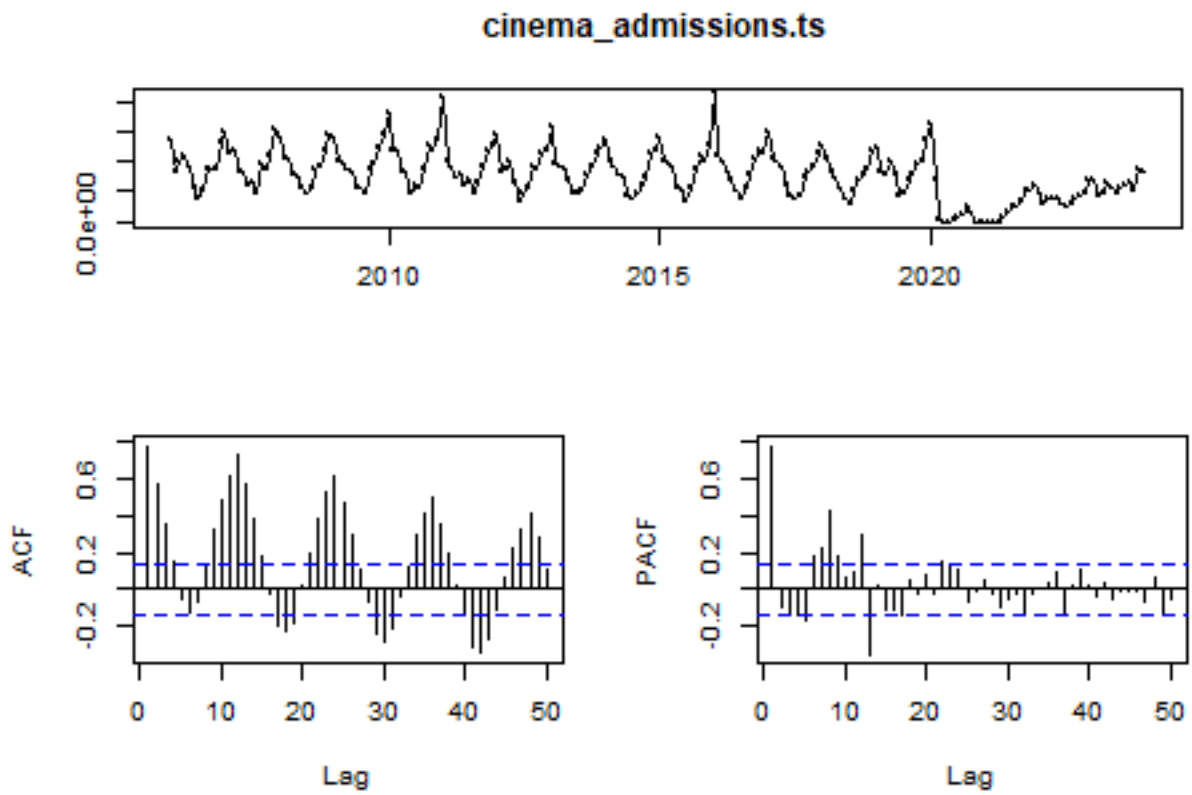
```r
min_max_scale <- function(x) {
    return((x - min(x, na.rm = TRUE))/(max(x, na.rm = TRUE) - min(x, na.rm = TRUE)))
}

# Apply Min-Max Scaling to the relevant columns
merged_ott_data$ott_subscriptions_scaled <- min_max_scale(merged_ott_data$ott_subscriptions_M)
merged_ott_data$ott_spending_scaled <- min_max_scale(merged_ott_data$ott_spending_M_EUR)
merged_ott_data$google_trend_ott_scaled <- min_max_scale(merged_ott_data$google_trend_ott)
# Convert Year to Date format for better plotting merged_ott_data$Year
# <- as.Date(paste0(merged_ott_data$Year, '-01-01'))
```

```r
# Compare trends
plot(merged_ott_data$Year, merged_ott_data$ott_subscriptions_scaled, type = "l",
    col = "blue", lwd = 2, xlab = "Year", ylab = "Normalized Value", main = "OTT Subscriptions, OTT Spe
lines(merged_ott_data$Year, merged_ott_data$ott_spending_scaled, col = "red",
    lwd = 2)
lines(merged_ott_data$Year, merged_ott_data$google_trend_ott_scaled, col = "green",
    lwd = 2)
legend("topleft", legend = c("OTT Subscriptions", "OTT Spending", "Google Trends OTT"),
    col = c("blue", "red", "green"), lwd = 2, bty = "n")
```

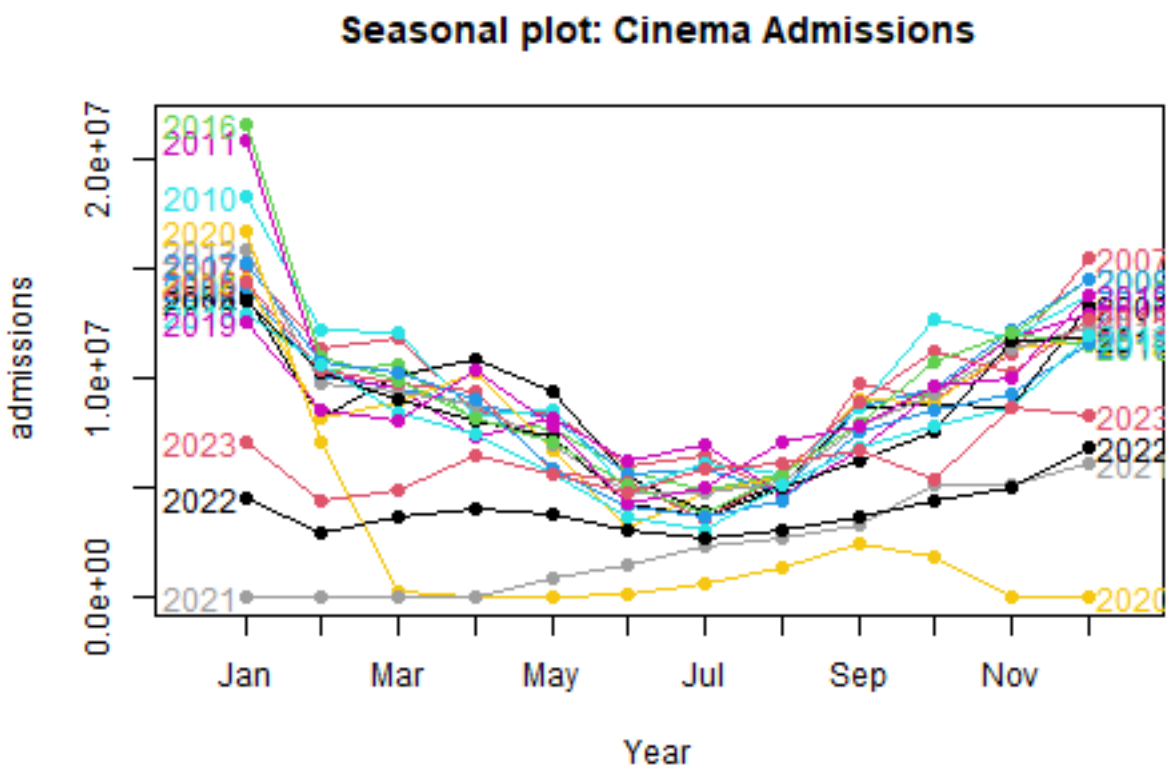


Time Series Decomposition

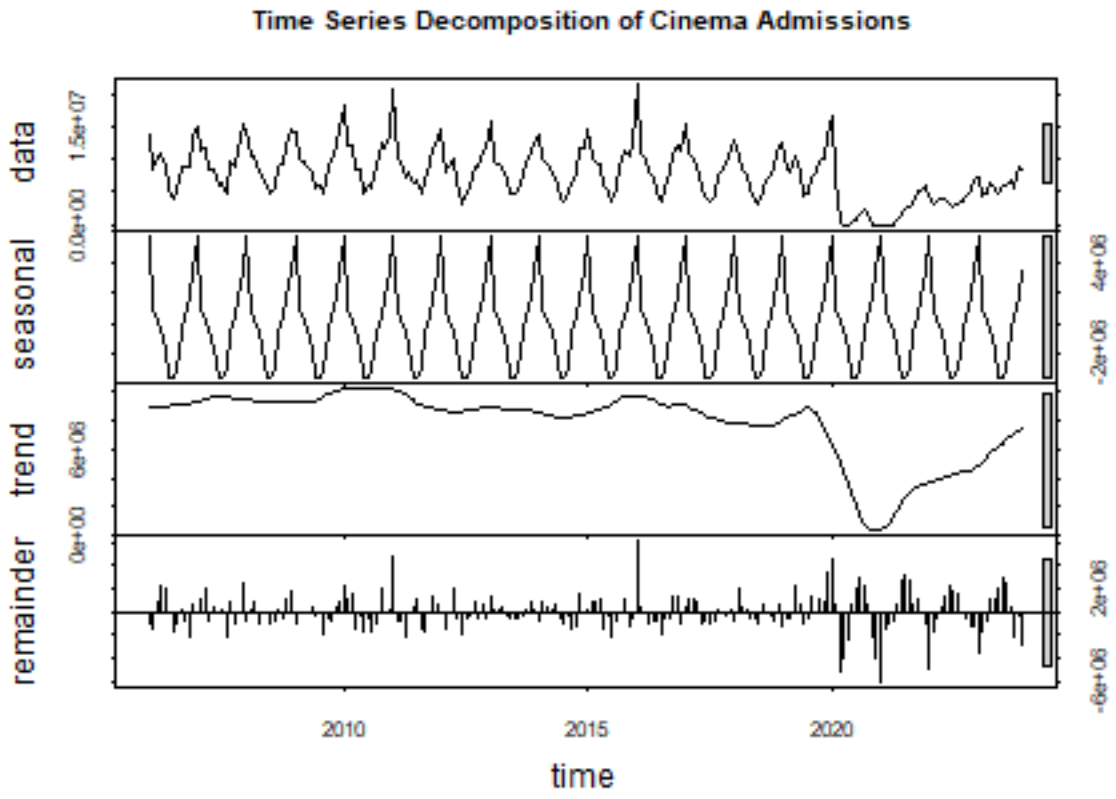
```
# Extract Cinema Admissions time series
cinema_admissions.ts <- ott_cinema_monthly.ts[, "cinema_admissions"]

# a general view of the data
tsdisplay(cinema_admissions.ts, lag.max = 50)
```




```
seasonplot(cinema_admissions.ts, ylab = "admissions", xlab = "Year", main = "Seasonal plot: Cinema Admi
    year.labels = T, year.labels.left = T, col = 1:20, pch = 19)
```




```r
# STL decomposition (Seasonal + Trend + Residual)
admissions_decomposed <- stl(cinema_admissions.ts, s.window = "periodic")

# Plot the decomposed components
plot(admissions_decomposed, main = "Time Series Decomposition of Cinema Admissions",
    cex.lab = 0.45, pch = 16, lty = 1, cex = 0.6)
```


Time Series Decomposition of Cinema Admissions
data
seasonal
trend
remainder
0.0e+00
1.5e+07
4e+06
-2e+06
6e+06
0e+00
2e+06
-6e+06
2010
2015
2020
time